

The Missing Rung: AI and the Entry-Level Tech Pipeline

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The Challenge

Generative AI is changing the economics of entry-level work. Tasks that once gave junior designers and engineers their first opportunity to learn—producing design assets, creating layouts and variations, writing boilerplate code, testing simple features, and producing first-draft interfaces—can increasingly be automated or completed with AI assistance. This creates a problem that is easy to miss when looking only at total employment: the technology profession can continue to exist while the entry point into the profession becomes weaker.

The immediate effect is on early-career workers. If a firm can obtain the output of a junior worker without hiring one, it has less reason to create an entry-level position. The longer-term effect falls on employers themselves: if fewer people enter the entry-level pipeline today, fewer people will emerge as experienced workers several years from now.

The question, therefore, is not simply whether AI reduces demand for junior workers. It is whether the market will correct the resulting training shortfall before the missing junior cohorts become a shortage of experienced talent.

This paper examines that question using four occupations with different expected exposure to AI automation: Graphic Designers, Software Developers, Web & Digital Interface Designers, and Home Health Aides. Software Developers provide a benchmark because early-career disruption is already well documented (Brynjolfsson et al., 2026). Graphic Designers provide the primary comparison, while Web & Digital Interface Designers provide a contrasting occupation in which AI is expected to augment more of the work. Home Health Aides provide a lower-exposure control.

The analysis is deliberately narrower than a claim that AI has caused all recent employment changes or will eliminate entire occupations. AI-exposure measures describe the mechanism being tested; employment data describe the observed labor-market pattern; and the pipeline model tests the timing implications under explicit assumptions.

The Economics

Three economic concepts explain why the entry-level pipeline can weaken even when firms are behaving rationally.

First, training creates a positive externality. When a firm hires and develops a junior worker, it bears the cost of training but does not necessarily capture the full return. The worker may eventually leave for another employer, allowing other firms to benefit from skills that the original employer helped develop. Each individual firm therefore has an incentive to invest less in training than would be optimal for the industry as a whole.

Generative AI can intensify that incentive. When AI makes routine junior tasks cheaper to perform, the immediate productivity benefit of hiring a junior worker falls, while the cost of training remains. The private business case for creating the entry-level position can therefore weaken even though the industry still needs the experienced workers those positions would eventually produce.

Second, the problem reflects short-run versus long-run elasticity. Firms can change junior hiring relatively quickly in response to new technology. The supply of experienced workers is less responsive in the short run because experience itself takes time to develop. The two sides of the labor market therefore adjust at different speeds. A firm can eliminate a junior position today, but it cannot create a fully experienced replacement tomorrow simply because demand for experienced workers has increased.

Third, the specificity rule suggests that the response should target the market failure rather than the technology causing it. If the underlying problem is underinvestment in training, the appropriate response is not to prevent firms from adopting AI. It is to restore investment in the development of workers who will form the future talent pool.

These concepts lead to a specific prediction: if AI reduces the need for junior production work, early-career hiring can fall before experienced-worker supply does. The resulting shortage may not appear until several years later, when the missing cohorts would have progressed through the pipeline.

Analysis

The first test asks whether the observed early-career employment pattern follows the predicted AI-exposure mechanism.

Using IPUMS Current Population Survey data (Flood et al., 2022), I compared employment among 22–25-year-olds in Graphic Designers and Web & Digital Interface Designers over a pooled baseline period of 2022 Q1–2023 Q4 and a current period of 2024 Q4–2026 Q3. Graphic Designers declined 38.3%, while Web & Digital Interface Designers declined 51.3%. The resulting Graphic Designer minus Interface Designer difference was –13.0 percentage points, with a 95% confidence interval from –38.6 to +12.6 percentage points.

This result does not support the original hypothesis that Graphic Designers would experience the larger decline. The confidence interval includes zero, so the two occupations cannot be statistically distinguished using this comparison. More importantly, the point estimate runs in the opposite direction from the prediction. The small Web & Digital Interface Designer sample also makes the estimate imprecise.

The result is therefore best treated as inconclusive rather than as evidence against AI disruption generally. The comparison does not establish that differences in AI automatability explain the employment changes between the two occupations. This is an important limitation because the mechanism test is the part of the analysis that most directly links AI exposure to the observed employment pattern.

The control also introduces caution. The CPS-based Home Health Aide series declined 22.3% over the same period. That is not consistent with the original expectation of a roughly flat control and means the analysis cannot rule out broader labor-market or occupation-specific forces as contributors to the observed decline. The within-occupation comparison of early-career and experienced workers therefore remains important: if experienced workers are also declining, the evidence for an entry-level-specific AI effect weakens.

A same-source CPS check on Software Developers, the occupation most directly cited as the primary study's clearest case of early-career disruption (Brynjolfsson et al., 2026), shows a smaller decline than that framing implies: employment among 22–25-year-old Software Developers in this CPS pull declined only about 1.6% from the baseline to the current period, with large quarter-to-quarter swings in both directions. This is notably smaller than the roughly 18% drop reported in the primary study's published series for the same occupation and age group over a comparable window. The two estimates are based on different data sources and methodologies—CPS survey data versus ADP payroll administrative data—which may contribute to the difference, but this paper does not attempt to reconcile the estimates. The discrepancy is reported rather than smoothed over, consistent with the paper's treatment of other results that did not match its initial expectations.

The second test addresses the question the first test cannot answer: what happens if reduced junior hiring persists, regardless of exactly how much of the initial decline is attributable to AI?

A cohort model tracks junior hires into the future mid-level workforce. The primary case assumes three years for a junior worker to progress to mid-level, with a five-year transition tested as a sensitivity case. Under the Depressed scenario, junior hiring remains at its current reduced level through 2030. Under the Recovery scenario, junior hiring returns to its pre-2022 trend by 2028.

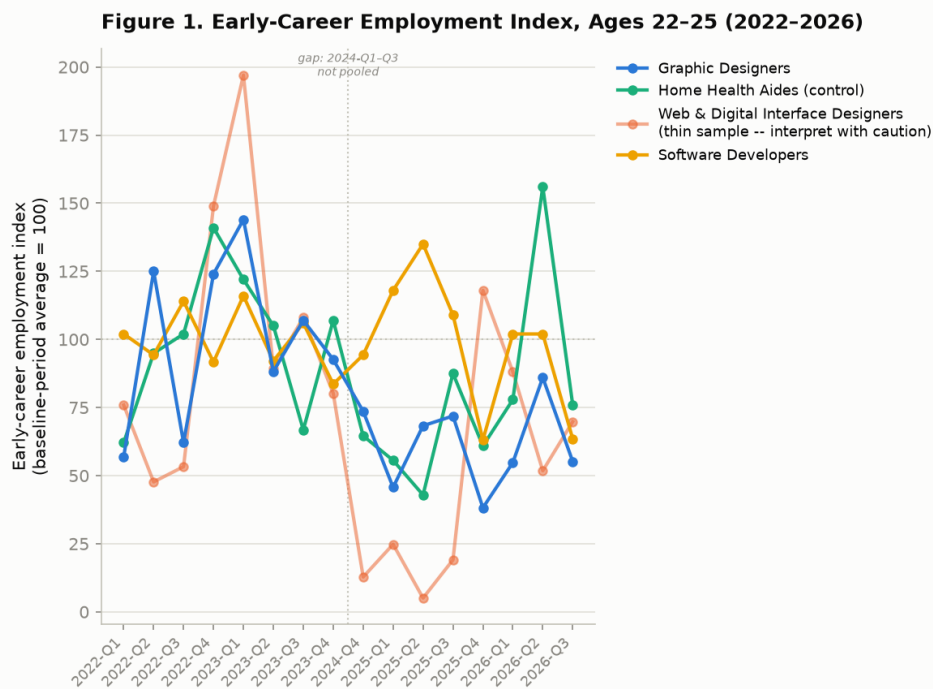
The timing result supports the second hypothesis. Under the Depressed scenario, the modeled mid-level shortage begins in 2026 with a three-year transition and in 2028 with a five-year transition. Both occur before 2030. Under the Recovery scenario, the gap narrows but remains open through 2035. The modeled gap falls from a peak of approximately 9,972 workers to approximately 7,716 under the three-year assumption and approximately 8,550 under the five-year assumption.

These numbers are not a forecast of the actual number of technology workers the economy will lack. They are the output of a scenario model built around an explicit training-

lag assumption. Their significance is the timing relationship: even if firms eventually recognize the shortage and restore junior hiring, those new hires cannot immediately become the experienced workers the earlier hiring reductions failed to produce. The result is therefore conditional rather than confirmatory: if depressed junior hiring persists, the pipeline cannot replenish experienced talent quickly enough, even if hiring eventually recovers.

Taken together, the two tests produce a more nuanced conclusion than the original hypothesis. The evidence does not establish that Graphic Designers have experienced a larger AI-driven decline than Interface Designers. The HHA control also prevents a clean interpretation of the employment decline as uniquely AI-driven. The pipeline model does not establish that Graphic Designers are experiencing the depressed-hiring scenario. Instead, it takes the broader early-career employment effect identified in the primary study (Brynjolfsson et al., 2026) as an input and asks what follows if that effect persists in an AI-exposed occupation. H2 therefore tests the timing of pipeline adjustment, not the occupation-specific causal mechanism tested by H1.

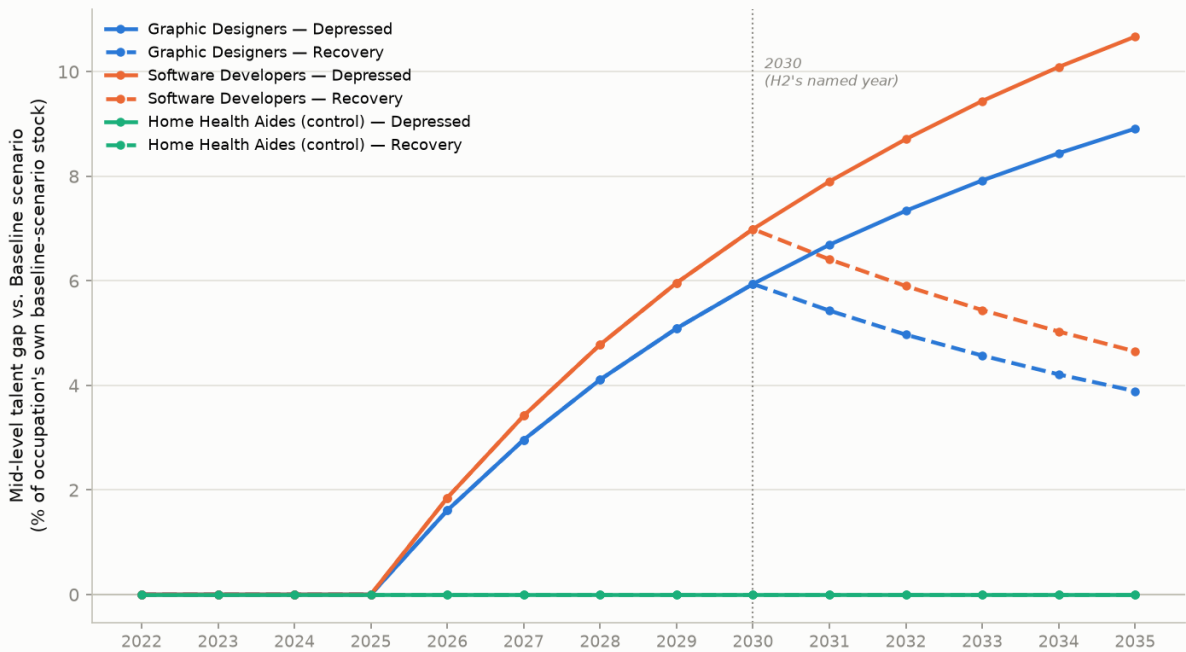
The central issue is therefore not whether the market eventually adjusts. It is whether it adjusts quickly enough.



Note. All four series are calculated from IPUMS CPS (employed respondents, EMPSTAT 10/12, ages 22-25), weighted and indexed to each occupation's own 2022Q1-2023Q4 baseline average. Interface Designers' quarterly cells are thin (as low as N=1-3 respondents in some current-period quarters); treat its quarter-to-quarter swings with caution relative to the other three series.

Figure 1. Early-career employment index, ages 22–25, 2022–2026. All four series are calculated from IPUMS CPS data (employed respondents, ages 22–25) using the same methodology, weighted and indexed to each occupation's own baseline-period average. Web & Digital Interface Designers' quarterly cells are thin (as low as N=1–3 respondents in some current-period quarters); its quarter-to-quarter swings should be read with more caution than the other three series.

Figure 2. Projected Mid-Level Talent Gap, Three Occupations (N=3, 2022-2035)



Note. Output of the cohort model (training lag N=3, the primary case) run separately for each occupation's own real inputs -- see capabilities/economic-research/pipeline-model-spec.md. Home Health Aides' Depressed_Hiring_Pct (1.0, no change) is inferred from the primary paper treating the least-exposed quintile as the omitted regression reference category, not read directly off a table cell -- flagged as an assumption, not a confirmed figure. Scenario outputs under explicit assumptions, not a forecast.

Figure 2 (appendix candidate). Projected mid-level talent gap, three occupations, 2022–2035. Output of the cohort model (training lag N=3, the primary case) run separately for each occupation's own real inputs. Gap is expressed as a percentage of each occupation's own baseline-scenario stock. Home Health Aides' Depressed_Hiring_Pct (1.0, no change) is inferred from the primary paper treating the least-exposed quintile as the omitted regression reference category, not read directly off a table cell — flagged as an assumption, not a confirmed figure.

Policy Recommendation

The case for intervention does not rest on H1 establishing that AI caused the Graphic Designer decline. H1 is inconclusive, and the decline in the Home Health Aide control means broader labor-market forces cannot be ruled out. The externality argument instead rests on the broader evidence in the primary study (Brynjolfsson et al., 2026) that AI-exposed occupations experienced disproportionate early-career employment declines, combined with the economic logic that firms may underinvest in training when the benefits of developing junior workers are not fully captured by the firm that pays for it. The policy implication is therefore conditional: where AI-driven reductions in junior work are creating a training shortfall, a

collective mechanism can address the resulting underinvestment more directly than attempting to slow AI adoption.

The most targeted response is therefore collective investment in early-career talent, rather than restrictions on AI adoption.

First, establish a mandatory sector-wide training levy. Firms within the defined sector would contribute to a shared fund for early-career training, with the obligation established through legislation or regulation rather than voluntary association membership. This addresses the free-rider problem: firms could not avoid contributing while benefiting from the experienced workers developed through other firms' investments. The fund could support shared training infrastructure and early-career development, while firms and the sector body retain responsibility for identifying workforce needs and coordinating delivery.

Second, policymakers should provide the legal foundation and oversight rather than direct day-to-day workforce decisions. Government would establish and enforce the funding mechanism through an appropriate statutory or regulatory framework. Firms and the sector body would remain responsible for determining training priorities and delivery. This keeps the policy targeted at the market failure rather than turning government into the manager of industry training.

Third, the levy should be complemented by targeted apprenticeship funding or entry-level hiring credits. A shared levy addresses the industry's longer-term underinvestment problem, while hiring support reduces the immediate cost and risk that can cause an individual employer to avoid hiring a junior worker.

Finally, firms should redesign junior roles around AI oversight, judgment, and development. If AI has removed some of the production work through which juniors traditionally learned, simply preserving the old job description is not enough. Junior workers can instead be given responsibility for directing, evaluating, testing, and correcting AI-generated work while developing the contextual knowledge and professional judgment

required for more advanced roles. The objective is to preserve the developmental function of the first rung even as the work performed on that rung changes.

The strongest objection is not only that the market may eventually self-correct, but that the observed employment decline may not be sufficiently AI-specific to justify intervention. The evidence here cannot rule out that possibility. That is why the recommendation should not be interpreted as a claim that every decline in junior employment warrants a training levy. Its justification depends on the broader AI-exposure evidence and on evidence that the loss of junior work is also a loss of training capacity. Where those conditions are present, waiting for a mid-level shortage to trigger new hiring still creates a timing problem: newly hired juniors require years to become experienced workers.

The appropriate intervention is consequently not to stop the market from adjusting, but to help it invest earlier. A collective training mechanism, combined with targeted incentives and redesigned junior roles, can preserve the entry-level pathway while allowing firms to capture the productivity benefits of AI.

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